Lakshya JEE 2.0(2024)

Indefinite Integration

DPP-02

1. $\int \frac{2x-1}{x-2}dx$

2. $\int \frac{\sin 2x+\sin 5x-\sin 3x}{\cos x+1-2\sin^2 2x}dx$

3. $\int \frac{\cos^4 x-\sin^4 x}{\sqrt{1+\cos 4x}}dx (\cos 2x>0)$

4. $\int \cos x^\circ dx$

5. $\int 4\cos\frac{x}{2}.\cos x.\sin\frac{21}{2}x\,dx$

6. $\int \frac{\cos 2x}{\cos^2 x\sin^2 x}dx$

7. $\int (3\sin x\cos^2 x-\sin^3 x)dx$

8. $\int \frac{\cos x-\sin x}{\cos x+\sin x}(2+2\sin 2x)dx$

9. $\int \frac{(1+x)^2}{x(1+x^2)}dx$

10. $\int \frac{\sec^2 x}{1+\tan x}dx$

11. $\int \frac{\sin(\ell nx)}{x}dx$

12. $\int \frac{x^6-1}{x^2+1}dx$

13. $\int \frac{\sin^3 x+\cos^3 x}{\sin^2 x\cos^2 x}dx$

14. $\int \frac{x^4+x^2+1}{2(1+x^2)}dx$

15. $\int \sqrt{1-\sin 2x}\,dx$

16. $\int \tan^{-1}\left(\frac{1+\cos x}{\sin x}\right)dx$

17. $\int \left[\sin^2\left(\frac{9\pi}{8}+\frac{x}{4}\right)-\sin^2\left(\frac{7\pi}{8}+\frac{x}{4}\right)\right]dx$

18. $\int \frac{\cos 8x-\cos 7x}{1+2\cos 5x}dx$

19. $\int (\tan^3 x-x\tan^2 x)dx$

20. $\int x.\sqrt{\frac{2\sin(x^2+1)-\sin 2(x^2+1)}{2\sin(x^2+1)+\sin 2(x^2+1)}}dx$

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. $2x + 3\ln(x - 2) + C$

2. $-2\cos x + C$

3. $\dfrac{x}{\sqrt{2}} + C$

4. $\dfrac{180}{\pi}\sin x^\circ + C$

5. $-\left[\dfrac{1}{9}\cos 9x + \dfrac{1}{10}\cos 10x + \dfrac{1}{11}\cos 11x + \dfrac{1}{12}\cos 12x\right] + C$

6. $-(\cot x + \tan x) + C$

7. $-\dfrac{\cos 3x}{3} + C$

8. $\sin 2x + C$

9. $\ln x + 2\tan^{-1}x + C$

10. $\ell n|1 + \tan x| + C$

11. $-\cos(\ell n x) + C$

12. $\dfrac{x^5}{5} - \dfrac{x^3}{3} + x - 2\tan^{-1}x + C$

13. $\sec x - \operatorname{cosec} x + C$

14. $\dfrac{1}{2}\left[\dfrac{x^3}{3} + \tan^{-1}x\right] + C$

15. $(\sin x + \cos x)\,sgn\,(\cos x - \sin x) + C$

16. $\dfrac{\pi x}{2} - \dfrac{x^2}{4}$

17. $-\sqrt{2}\cos\dfrac{x}{2} + C$

18. $\dfrac{\sin 3x}{3} - \dfrac{\sin 2x}{2} + C$

19. $\dfrac{(\tan x - x)^2}{2} + C$

20. $\log\left|\sec\dfrac{(x^2+1)}{2}\right| + C$

PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if